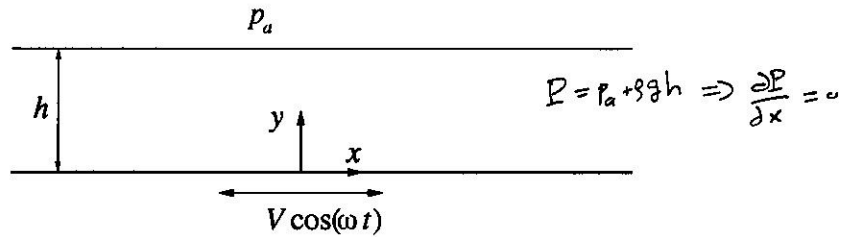


VF2

Consider the periodic motion induced in a liquid layer of thickness h when the bottom wall moves with velocity $V \cos(\omega t)$. Write the equation with boundary conditions that determines $v_x(y, t)$. Consider separately the cases $\omega h^2/\nu \gg 1$ and $\omega h^2/\nu \ll 1$. Obtain the general solution and show how it reduces to the limiting solutions studied earlier in the appropriate limits.



$$\frac{\partial v_x}{\partial t} = \nu \frac{\partial^2 v_x}{\partial y^2} \quad \begin{cases} y=0: v_x = V \cos(\omega t) \\ y=h: \frac{\partial v_x}{\partial y} = 0 \quad (\text{NEGLECTING VISCOSITY OF AIR}) \end{cases}$$

EXACT SOLUTION:

$$\frac{\partial v^*}{\partial t} = \nu \frac{\partial^2 v^*}{\partial y^2} \quad \begin{cases} y=0: v^* = V e^{i\omega t} \\ y=h: \frac{\partial v^*}{\partial y} = 0 \end{cases} \rightarrow v^* = V e^{i\omega t} f(y)$$

$$\bar{v} = \text{Re} \left[V e^{i\omega t} \frac{e^{+(1+i)\sqrt{\frac{\omega h^2}{2\nu}}(1-\frac{y}{h})} + e^{-(1+i)\sqrt{\frac{\omega h^2}{2\nu}}(1-\frac{y}{h})}}{e^{(1+i)\sqrt{\frac{\omega h^2}{2\nu}}} + e^{-(1+i)\sqrt{\frac{\omega h^2}{2\nu}}}} \right]$$

$i\omega f = f'' \quad \begin{cases} f(0)=1 \\ f'(h)=0 \end{cases}$

$f = c_1 e^{(1+i)\sqrt{\frac{\omega}{2\nu}}y} + c_2 e^{-(1+i)\sqrt{\frac{\omega}{2\nu}}y}$

$$\frac{\omega h^2}{2\nu} \rightarrow 0: \bar{v} \rightarrow \text{Re} [V e^{i\omega t}] = V \cos(\omega t)$$

$$\frac{\omega h^2}{2\nu} \rightarrow \infty: \bar{v} \rightarrow \text{Re} [V e^{i\omega t} e^{-(1+i)\sqrt{\frac{\omega}{2\nu}}y}] = V e^{-\sqrt{\frac{\omega}{2\nu}}y} \cos(\omega t - \sqrt{\frac{\omega}{2\nu}}y) \rightarrow \text{STOKES}$$

LIMITING SOLUTIONS

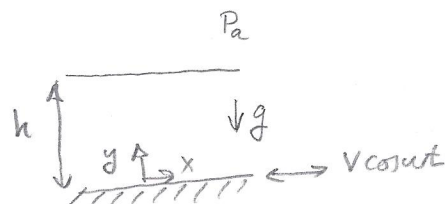
$$\frac{\partial v_x}{\partial t} = \nu \frac{\partial^2 v_x}{\partial y^2} \quad \begin{cases} y=0: v_x = V \cos(\omega t) \\ y=h: \frac{\partial v_x}{\partial y} = 0 \end{cases}$$

$\frac{V}{\omega^{-1}} \quad \nu \frac{V}{h^2}$

$$\frac{\omega h^2}{\nu} \ll 1 \rightarrow \text{Q-S solution} \quad \frac{\partial^2 v_x}{\partial y^2} = 0 \quad \begin{cases} y=0: v_x = V \cos(\omega t) \\ y=h: \frac{\partial v_x}{\partial y} = 0 \end{cases} \rightarrow \boxed{v_x = V \cos(\omega t)} \quad \text{THE LIQUID LAYER MOVES LIKE A SOLID!!}$$

$$\frac{\omega h^2}{\nu} \gg 1 \rightarrow \frac{\partial v_x}{\partial t} = 0 \quad \text{THE MOTION IS RESTRICTED TO A THIN LAYER OF CHARACTERISTIC THICKNESS } \sqrt{\nu/\omega} \ll h \rightarrow \frac{\partial v_x}{\partial t} = \nu \frac{\partial^2 v_x}{\partial y^2} \quad \begin{cases} y=0: v_x = V \cos(\omega t) \\ y \gg \sqrt{\nu/\omega} \rightarrow v_x \rightarrow 0 \end{cases}$$

$$\boxed{v_x = V e^{-\sqrt{\frac{\omega}{2\nu}}y} \cos(\omega t - \sqrt{\frac{\omega}{2\nu}}y)} \quad \text{STOKES!!}$$



MOMENTUM Y: $0 = -\frac{\partial P}{\partial y} \Rightarrow P = P_0 + \rho g y = P_0 + \rho g h = \text{const.}$

$\Downarrow$
 $\frac{\partial P}{\partial x} = 0$

LOOK FOR SOLUTION $\vec{v} = (u, v) = (u, 0)$, CONTINUITY $\Rightarrow \frac{\partial u}{\partial x} = 0 \Rightarrow u = u(y, t)$

MOMENTUM X: $\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial y^2}$

$y=0: u = V \cos \omega t$

$y=h: \frac{\partial u}{\partial y} = 0$

WE LOOK FOR PERMANENT SOLUTION $\Rightarrow$ NO INITIAL CONDITIONS NEEDED
OTHERWISE, WE COULD SET, AT THE WALL $V \sin \omega t$ and use
 $t=0: u=0$ as initial condition. (Compatible)

NOTICE THAT $V \cos \omega t = \text{Re}[V e^{i \omega t}] = \text{Re}[V \cos \omega t + i V \sin \omega t]$

THIS SUGGESTS TO USE COMPLEX VARIABLE, IN NON-DIMENSIONAL FORM

$\frac{u}{V} = e^{i z} f(z), \quad z = \omega t, \quad \bar{z} = \frac{y}{h} \Rightarrow V i \omega e^{i \omega t} f(z) = \frac{\nu}{h^2} V \frac{d^2 f}{dz^2}$

$y=0 \Rightarrow z=0: V e^{i z} f(0) = V e^{i z}$

$y=h \Rightarrow z=1: \frac{df}{dz} = 0$

SO THAT WE GET THE MATHEMATICAL NON-DIMENSIONAL PROBLEM:

$f'' - 2i\Omega^2 f = 0$ WITH $2\Omega^2 = \frac{\omega h^2}{\nu}$. 1 PARAMETER !!

$f(0) = 1$

$f'(1) = 0$

REMEMBER $u = V \cdot \text{Re}[e^{i z} f(z)]$. WE WILL FIND THE COMPLEX SOLUTION AND THEN THE REAL PART.

ODE, LINEAR $\rightarrow$ EIGENVALUES: $\lambda^2 - 2i\Omega^2 = 0 \rightarrow \lambda = \sqrt{2i\Omega^2} = \sqrt{2}\Omega e^{i\pi/4}$

TWO ROOTS: $\lambda_1 = \sqrt{2}\Omega e^{i\pi/4} = \Omega(1+i)$
 $\lambda_2 = \sqrt{2}\Omega e^{i3\pi/4} = -\Omega(1+i)$

GENERAL SOLUTION: $f = A e^{\Omega(1+i)z} + B e^{-\Omega(1+i)z}$, WITH A, B COMPLEX NUMBERS TO BE DETERMINED FROM BOUNDARY CONDITIONS.

$f(0) = 1 = A + B$
 $f'(1) = 0 = A \Omega(1+i) e^{\Omega(1+i)} - B \Omega(1+i) e^{-\Omega(1+i)}$ $\rightarrow \begin{cases} A+B=1 \\ A e^{\Omega(1+i)} - B e^{-\Omega(1+i)} = 0 \end{cases} \begin{matrix} (1) \\ (2) \end{matrix}$

(1) $\Rightarrow B = 1-A \xrightarrow{(2)} A e^{\Omega(1+i)} - (1-A) e^{-\Omega(1+i)} = 0 \Rightarrow A [e^{\Omega(1+i)} + e^{-\Omega(1+i)}] = e^{-\Omega(1+i)}$

SO THAT $A = \frac{e^{-\Omega(1+i)}}{e^{\Omega(1+i)} + e^{-\Omega(1+i)}} \Rightarrow B = 1 - \frac{e^{-\Omega(1+i)}}{e^{\Omega(1+i)} + e^{-\Omega(1+i)}} \Rightarrow B = \frac{e^{\Omega(1+i)}}{e^{\Omega(1+i)} + e^{-\Omega(1+i)}}$

GENERAL COMPLEX SOLUTION:

$\frac{u}{V} = \frac{e^{-\Omega(1+i)} e^{\Omega(1+i)z} + e^{\Omega(1+i)} e^{-\Omega(1+i)z}}{e^{\Omega(1+i)} + e^{-\Omega(1+i)}} \cdot e^{i \omega t}$

TO FIND THE REAL PART, MULTIPLY BY CONJUGATE OF DENOMINATOR:

$(e^{\Omega(1+i)} + e^{-\Omega(1+i)}) \cdot (e^{\Omega(1-i)} + e^{-\Omega(1-i)}) = e^{2\Omega} + e^{2\Omega i} + e^{-2\Omega i} + e^{-2\Omega} = e^{2\Omega} + e^{-2\Omega} + 2\cos 2\Omega$ (REAL).

So THAT.

$$\frac{u}{v} = \frac{e^{-\Omega(1+i)z} e^{\Omega(1+i)z} (e^{\Omega(1-i)} + e^{-\Omega(1-i)}) + e^{\Omega(1+i)z} e^{-\Omega(1+i)z} (e^{\Omega(1-i)} + e^{-\Omega(1-i)})}{e^{2\Omega} + e^{-2\Omega} + e^{-2\Omega i} + e^{2\Omega i}} \cdot e^{i\omega t}$$

$$= \frac{e^{\Omega(1+i)z} [e^{-2\Omega} + e^{-2\Omega i}] + e^{-\Omega(1+i)z} [e^{2\Omega} + e^{2\Omega i}]}{e^{2\Omega} + e^{-2\Omega} + e^{-2\Omega i} + e^{2\Omega i}} \cdot e^{i\omega t}$$

$$= \frac{e^{\Omega z [1 + i(1 + \frac{\omega t}{\Omega z})]} (e^{-2\Omega} + e^{-2\Omega i}) + e^{-\Omega z [1 + i(1 - \frac{\omega t}{\Omega z})]} (e^{2\Omega} + e^{2\Omega i})}{e^{2\Omega} + e^{-2\Omega} + e^{-2\Omega i} + e^{2\Omega i}}$$

$$= \frac{e^{\Omega z} e^{i(\Omega z + \omega t)} (e^{-2\Omega} + e^{-2\Omega i}) + e^{-\Omega z} e^{-i(\Omega z - \omega t)} (e^{2\Omega} + e^{2\Omega i})}{e^{2\Omega} + e^{-2\Omega} + e^{-2\Omega i} + e^{2\Omega i}}$$

$$\frac{u}{v} = \text{Re}[J] = \frac{1}{e^{2\Omega} + e^{-2\Omega} + 2\cos 2\Omega} \left[e^{\Omega z} e^{-2\Omega} \cos(\Omega z + \omega t) + e^{\Omega z} \cos(\Omega z + \omega t - 2\Omega) + e^{-\Omega z} e^{2\Omega} \cos(\Omega z - \omega t) + e^{-\Omega z} \cos(2\Omega - \Omega z + \omega t) \right]$$

LIMITING CASES

① $\Omega \gg 1$

GENERAL SOLUTION $\frac{u}{V} \approx \frac{e^{2\Omega} e^{-\Omega z} \cos(\Omega z - \omega t) + \dots}{e^{2\Omega} + \dots} \approx e^{-\Omega z} \cos(\Omega z - \omega t)$

IN PHYSICAL VARIABLES: $\frac{u}{V} \approx e^{-\sqrt{\frac{\omega h^2}{2\nu}} \cdot \frac{y}{h}} \cos\left(\sqrt{\frac{\omega h^2}{2\nu}} \cdot \frac{y}{h} - \omega t\right) = e^{-\frac{y}{\sqrt{2\nu/\omega}}} \cdot \cos\left(\omega t - \frac{y}{\sqrt{2\nu/\omega}}\right)$
 h DISAPPEARS!!!

THIS COULD BE ANTICIPATED FROM THE ODE PROBLEM

$\Omega \gg 1 \Rightarrow \begin{cases} f'' - 2i\Omega^2 f = 0 \\ f(0) = 1 \\ f'(h) = 0 \end{cases} \rightarrow f = 0$ EXCEPT BOUNDARY LAYER NEAR WALL.

IN DIMENSIONAL VARIABLES: $\begin{cases} \frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial y^2} \\ u(0) = V \cos \omega t \\ u'(h) = 0 \end{cases} \left\{ \begin{array}{l} \omega \nu \sim \nu \frac{V}{\delta^2} \rightarrow \delta \sim \sqrt{\frac{\nu}{\omega}} \\ \frac{h}{\delta} = \sqrt{\frac{h^2 \omega}{\nu}} \gg 1 \text{ IF } \Omega \gg 1. \end{array} \right.$

$u = 0$ EVERYWHERE EXCEPT IN LAYER OF THICKNESS δ

$\frac{y}{\delta} = \eta$; $\frac{u}{V} = e^{i\omega t} f(\eta) \rightarrow i\omega \frac{u}{V} = \nu \frac{\partial^2 u}{\partial y^2} \rightarrow \frac{\nu}{\delta^2} f'' = i\omega f$
 $\begin{cases} f'' - i f = 0 \\ f(0) = 1 \\ f'(\infty) = 0 \end{cases} \quad \lambda^2 - i = 0 \rightarrow \lambda = \begin{cases} \frac{1+i}{\sqrt{2}} \\ -\frac{1+i}{\sqrt{2}} \end{cases}$

$f = A \cdot e^{\frac{1+i}{\sqrt{2}} \eta} + B e^{-\frac{1+i}{\sqrt{2}} \eta}$

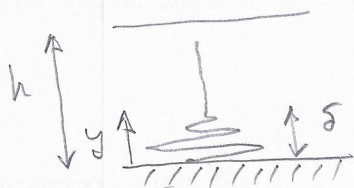
$f(0) = 1 = A + B$

$f'(\infty) = 0 = A \frac{1+i}{\sqrt{2}} e^{\frac{\eta}{\sqrt{2}}} e^{\frac{i\eta}{\sqrt{2}}} - B \frac{1+i}{\sqrt{2}} e^{-\frac{\eta}{\sqrt{2}}} e^{-\frac{i\eta}{\sqrt{2}}}$

∞ UNLESS $A = 0 \rightarrow A = 0$
 $B = 1$

$\frac{u}{V} = e^{-\frac{\eta}{\sqrt{2}}} e^{-\frac{i\eta}{\sqrt{2}}} e^{i\omega t} = e^{-\frac{y}{\sqrt{2}}} e^{-i\left(\omega t - \frac{y}{\sqrt{2}}\right)}$

REAL PART: $\frac{u}{V} = e^{-\frac{y}{\sqrt{2\nu/\omega}}} \cdot \cos\left(\omega t - \frac{y}{\sqrt{2\nu/\omega}}\right)$ SAME RESULT!!!



② $\Omega \ll 1$

GENERAL SOLUTION:

$$\frac{u}{v} \rightarrow \frac{1}{4} [\cos \omega t + \cos \omega t + \cos \omega t + \cos \omega t] \approx \cos \omega t$$

THIS COULD BE ANTICIPATED FROM THE ODE:

$$\left. \begin{aligned} f'' - i\Omega f &= 0 \rightarrow f'' \approx 0 \\ f(0) &= 1 \\ f'(1) &= 0 \end{aligned} \right\} \text{ SOLUTION: } f \approx 1 \Rightarrow \frac{u}{v} = 1 \cdot e^{i\omega t}$$

REAL PART: $\frac{u}{v} = \cos \omega t$

SAME RESULT.!!